

D4 Final Design Review

Presentation Structure (Draft)

The FDR should tell the three connected stories:

1	What we were asked to do	Task intro, environment, equipment, project split into 4 workstreams/parts
2	What we built & how it works	System overview, 4 workstreams/parts, requirements verification, step-by-step demo walkthrough
3	How we worked as a team	Roles, communication, leadership, tools

Slide Distribution

1 Slide	Title Slide
2 Slides	Introduction to the Task (search and rescue scenario) Member: Herish
5 Slides	Overview of the System & 4 Workstreams/parts All members (each of us will describe our part of the project 1 slide per each member) Option 1 Edward: data collection and vision training Demetro: SIM and Vision Herish: PLB path Ziyan: geofence and main path planning Robin: machine states and main brain of the code Option 2 Two members presents only
1-2 Slides	How the System Meets the Requirements Member: Ziyan
2-3 Slides	How the Implemented Design Works (Step by Step) Robin
2 Slides	How the Team Was Built Edward
1 Slide	Closing Summary & Demo Day roles (e.g. safety pilot, Ground station operator, camera feed monitor, Mission manager, Payload and first Aid) Demetro

Slide 1 : Title Slide

NO presenter

What to include:

- Project/mission name(in a box maybe) (e.g. "AENGM0074 Search & Rescue UAV System")
- Team/company name
- Date and presentation type (D4 Final Design Review)
- All 5 team member names
- University of Bristol logo

Design tip:

The first thing external guests will see. Less is more here.

Section 1: Introduction to the Task (max 3 slides)

Goal: Make sure any guest can understand(tell a story) what we were asked to do, what success looks like, and what constraints we had to work within.

What to cover (these are just ideas so be creative):

- What is the mission? Introduce the SAR scenario simply (lost hiker, wilderness environment, first aid delivery)
- What does a successful end-to-end mission look like?
- The environment: Fenswood Wilderness ,size, boundaries, SSSI no-fly zone, take-off location
- Key constraints: 50m altitude limit, geofences, PLB activation mid-flight,...
- Equipment we were given: drone platform (Hexsoon), Raspberry Pi 5, camera, payload release mechanism -> don't go into details because they system overview will do (maybe show pics here)

Guiding questions to answer:

- What problem were we asked to solve. In one or two plain sentences?
- What does the flight area look like and what are its boundaries? (map or diagram maybeee)
- What hardware were we given to work with and what are its key capabilities and limitations?
- What made this task challenging? (e.g. PLB only activates 5–15 min in so we don't know where they dummy is, no visible field markers, restriction)
- What does 'mission complete' look like and what would we need to demonstrate to fully satisfy the brief?

Suggested visuals:

- Annotated satellite image of Fenswood, with flight area, SSSI, TOL, and survey area labelled
- Photo or diagram of the drone platform with key components labelled
- One-sentence mission statement displayed prominently

Section 2: Overview of the System(max 6 slides)

Goal: Show what we built, how the 4 parts of the projec connect, and why we made the design decisions we did.

Introduce these 4 workstreams/parts below clearly:

Machine States & Logic	The brain of the system. Controls what the drone decides to do and when state transitions, mission phases, event handling (e.g. PLB trigger)
Computer Vision	How the drone sees and identifies items of interest and the casualty. Algorithm selection, testing process, detection accuracy
Path Planning	How the drone decides where to fly (search patterns, waypoints, search area, focus area and what algorithms were used)
Hardware	The physical platform , drone, flight controller, companion computer, camera, payload release.... Not to detailed

Guiding questions to answer:

- What does the full system architecture look like? (use a block diagram or flow chart, show all 4 workstreams/parts and how they connect)
- What level of autonomy did we choose and why? The decision needs justification. Maybe we should refer to the rubrics and marking criteria to justify our decisions
- For the Computer Vision process: what was our process of exploration and testing? What did we try, what didn't work, and what did we settle on and why?
- What were the biggest design trade-offs we made? (e.g. autonomy vs. reliability, detection speed vs. accuracy) This is were we can be smart and talk about real world and clients needs and the practicality of the design. Can also talk about features Vs. benefits (industry people love this)
- How does the ground station interface work? Can you show a screenshot?-> I think robin will already cover this but just to keep it as a note
- How do the 4 workstreams/parts talk to each other? where are the key handoffs? (e.g. Vision tells Logic when it detects something, Logic tells Path Planning to reroute) This will also be in the Robins part but

Suggested visuals:

- System architecture block diagram showing all 4 workstreams/parts and their interfaces
- Screenshot of the ground station UI and GUI, SIM
- Example detections from the vision system (Dmytro provide all your pics)
-

Section 3: How the System Meets the Requirements(max 2 slides)

Goal: Show the audience that every requirement has been addressed and evidenced(but we should be honest).

Requirements reference. Map each one to the system:

Req	What it covers	Which workstream/parts owns it
R01	Drone stays within Flight Area at all times	Hardware / Path Planning
R02	Drone does not fly over the SSSI	
R03	Take-off within 5m of	
R04	Max altitude 50m	
R05	Search the Search Area and identify items of interest	Vision

Guiding questions to answer:

- what specific part of our system satisfies which requirements ?
- What evidence do we have? (flight logs, test data, screenshots, detection results, photos)
- Is there anything we partially meet or are still working towards? Is there anything we didn't meet ?
- What uncertainties remain going into demo day?

Suggested visuals:

- A simple table maybe
- screenshots or anything else

Section 4: How the Implemented Design Works(max 4 slides)

Goal: Help the audience follow along on demo day. By the end of this section, everyone in the room should know exactly what the drone is doing at every moment in the air.

Walk through the mission step by step:

items
Pre-flight & takeoff
Search pattern begins
Items of interest detected
PLB activates (how will it be provided)
Focus Area search
Detection , coordination with Demetro the vision monitor
Approach & landing
Kit deployment
Return to Home

Guiding questions to answer:

- Can we show this as a flow diagram or annotated mission timeline rather than just bullet points?
- When the PLB activates mid-flight — exactly what happens in the system? Does the current search pause, abort, or complete?
- How does the drone land precisely? If it does
- What failsafe are in place and when do they activate? (RTH, emergency)
- and what is your contingency plan or alternatives?

Suggested visuals:

This slide is our preview of demo day. After this, the audience should be able to watch the flight and know exactly what is happening. Any pictures and images will be good

Section 5: How the Team Was Built(max 3 slides)

Goal: Show that this wasn't just technical and act as if we are a real engineering company with clear roles, structure, and practice.

Guiding questions to answer:

- How did we decide who owns which workstream/parts, skills-based, interest-based, volunteered(I doubt it haha)?

Machine States & Logic: who owned it?

Computer Vision: who owned it?

Path Planning: who owned it?

Hardware: who owned it?

- How did we communicate day-to-day? What tools did we use? (Notion,Trello, WhatsApp, Teams, GitHub, shared docs?)
- How did we run meetings, how often, how long, any formal structure?
- How did we manage tasks and deadlines
- How did the leadership worked-> maybe we can mention that the role of leadership was shared

Suggested visuals:

- Team org chart or I don't know anything you think is good
- Screenshot of our project management tool / communication channels maybe a screen shot of online meeting or notes or some fun pictures that should have we did our meetings like the example below

Remember: Don't just list what we did, reflect on why certain things worked and what we learned.



Last Slide: Closing Summary(max 2 slides)

Goal: Maybe a brief, clean, confident summary of what is ready to go now , I not sure about this slide. The idea to just quickly summarize the project and end it nicely

Share you idea about what can go into the closing

Demo day role :

- Safety pilot: herish
- Ground station operator (also the one who will talk during the mission)
- Camera Stream Monitor : demetro
- Payload & First Aid Technician, photography and documentation: Edward
- Mission manger: ziyen

These might change I don't know but the reason I choose them like that is because those who already tested the drone and has been present for the last two flight days have already practiced the mission and it will be hard for other to do this job during the demo day

Closing statement or a quote:

| *"Something nice be written here "*

General Tips :

- We only have 15 min to present
- Don't put too much text
- Practice talking through what you write and see how it sounds
- Know who goes before you and after you and make the transition smoothly
- Share all the pictures you have taken during the course so we can use them for the presentation and report
- Ask if you don't know what is expected from you
- This is easy task and you have to prepare max 3 slides if very necessary maybe more but that is it for now
- I will need everyone's parts and make it look like one presentation so please don't use any crazy designs or figures that can't be re used or can't fit in the slides. Simply, think about the integration of everyone's work
- Talk about the most important thing
- Don't go so deep into the technical parts unless necessary
- If you need data, pictures, answers for your slides just ask one of us and maybe we can help. Don't assume answers

Disclaimer:

In some parts I have used AI to extend on my notes so if something is not clear just ask me